

WASP-177b

R-2476

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-177_250728 · created 2026-08-23T23:47:13+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460884.882 +/- 0.014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.048
Transit depth (Rp/Rs)^2	2.88 +/- 1.63 [%]
Orbital Inclination (inc)	85.32 +/- 2.86
Airmass coefficient 1 (a1)	1.11 +/- 0.48
Airmass coefficient 2 (a2)	0.1 +/- 0.97
Scatter in the residuals of the lightcurve fit is	45.9 %
Best Comparison Star	None
Optimal Aperture	5.94
Optimal Annulus	10.49
Transit Duration (day)	0.062 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.17 ± 0.048 vs 0.1618 (deviation 0.15σ)
Duration fit vs published	0.062 d vs 0.0721 d (deviation 0.26σ)
Mid-time O–C	30.8 min
Depth SNR	3.1
Residual scatter	45.9 %

Light curve

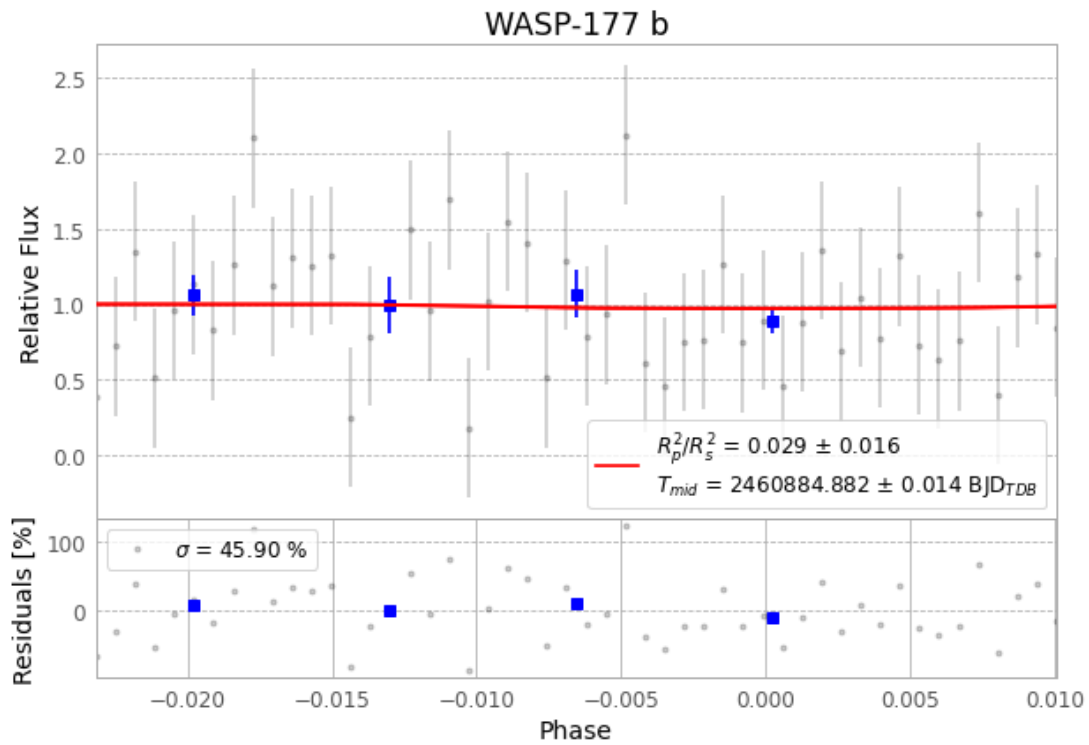


Figure 1 — FinalLightCurve_WASP-177 b_2025-07-28.png (SHA-256 72ccb8fd60cb82e2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [339, 42], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 570b5b7a52ede9ee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

50 FITS frames (33 MB), WASP-177250728071815.FITS → WASP-177250728094515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-177-250728.log (95b1c1c60254...), FinalLightCurve_WASP-177 b_2025-07-28.png (72ccb8fd60cb...), FinalLightCurve_WASP-177 b_2025-07-28.csv (eb2713fbbb87...)

Compute resources

Wall time 174.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 23:47Z.